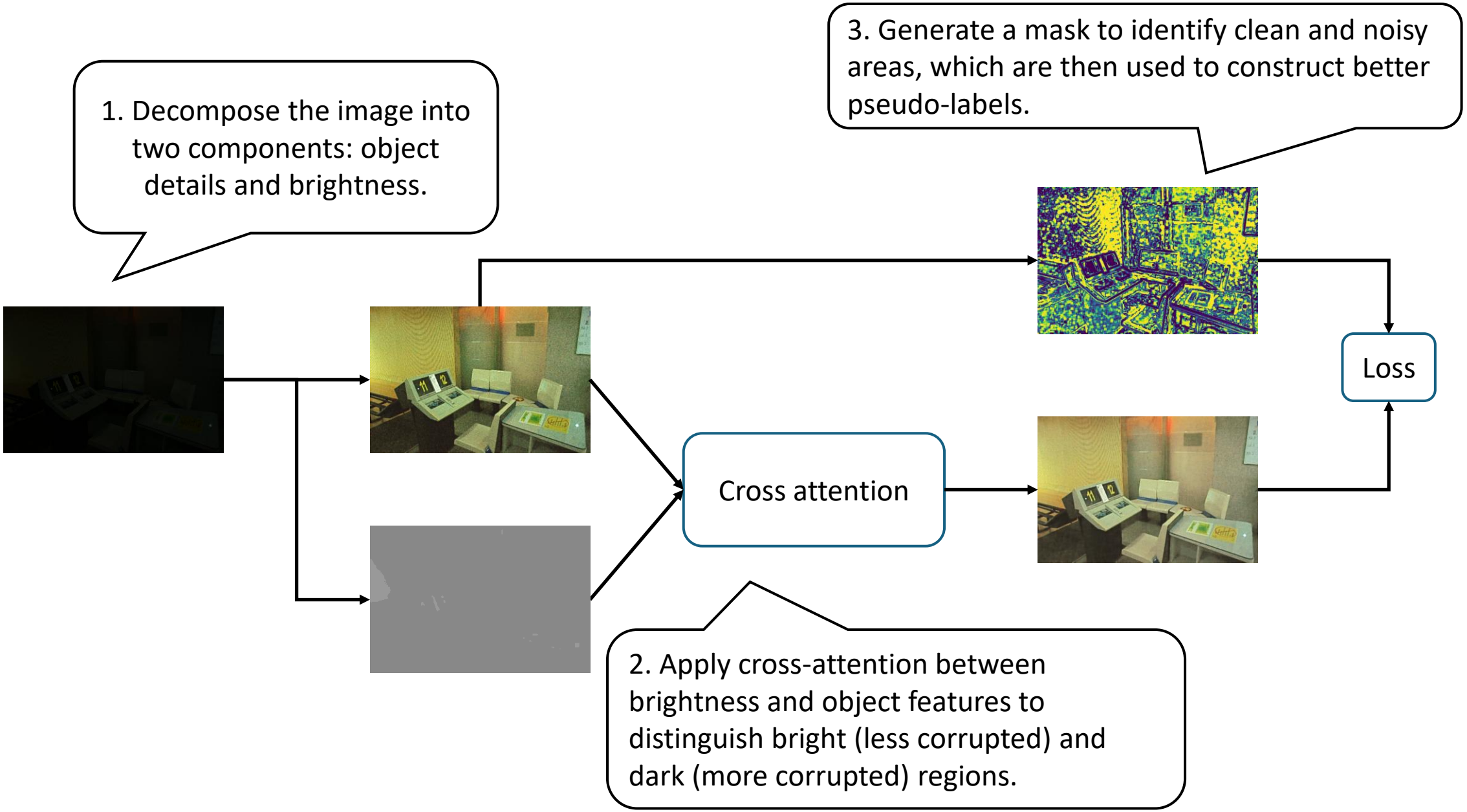
Image Enhancement

- Degradations can occur anywhere: noise, blur, low light, or even complex distortions.
- Goal: Restore images to reveal richer information.

Top: degraded images. Bottom: high-quality images.

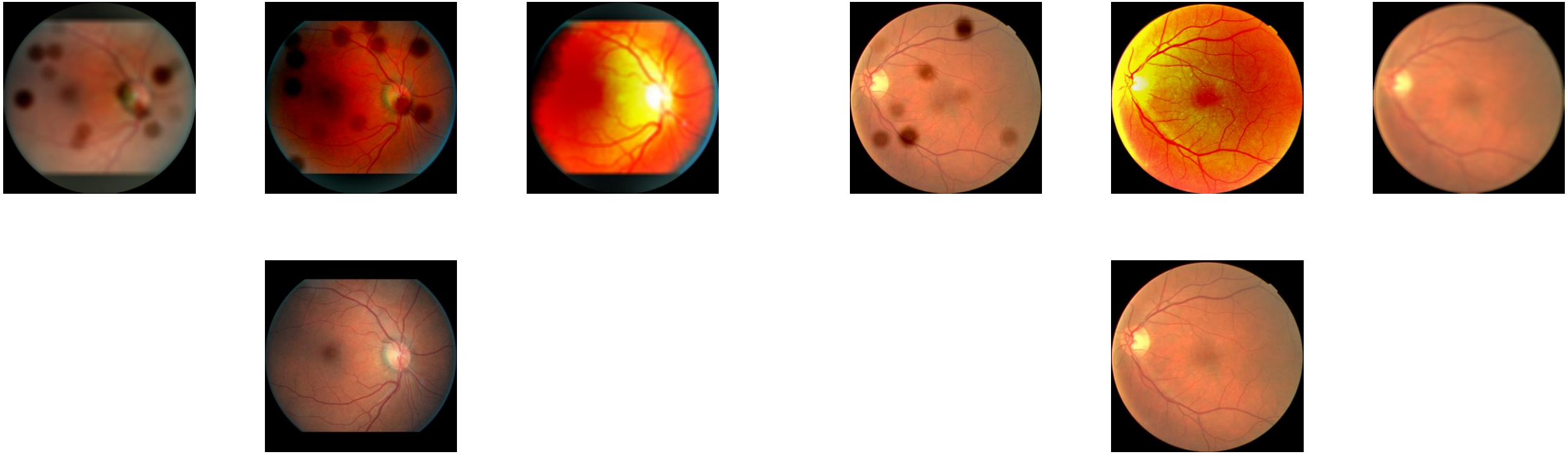
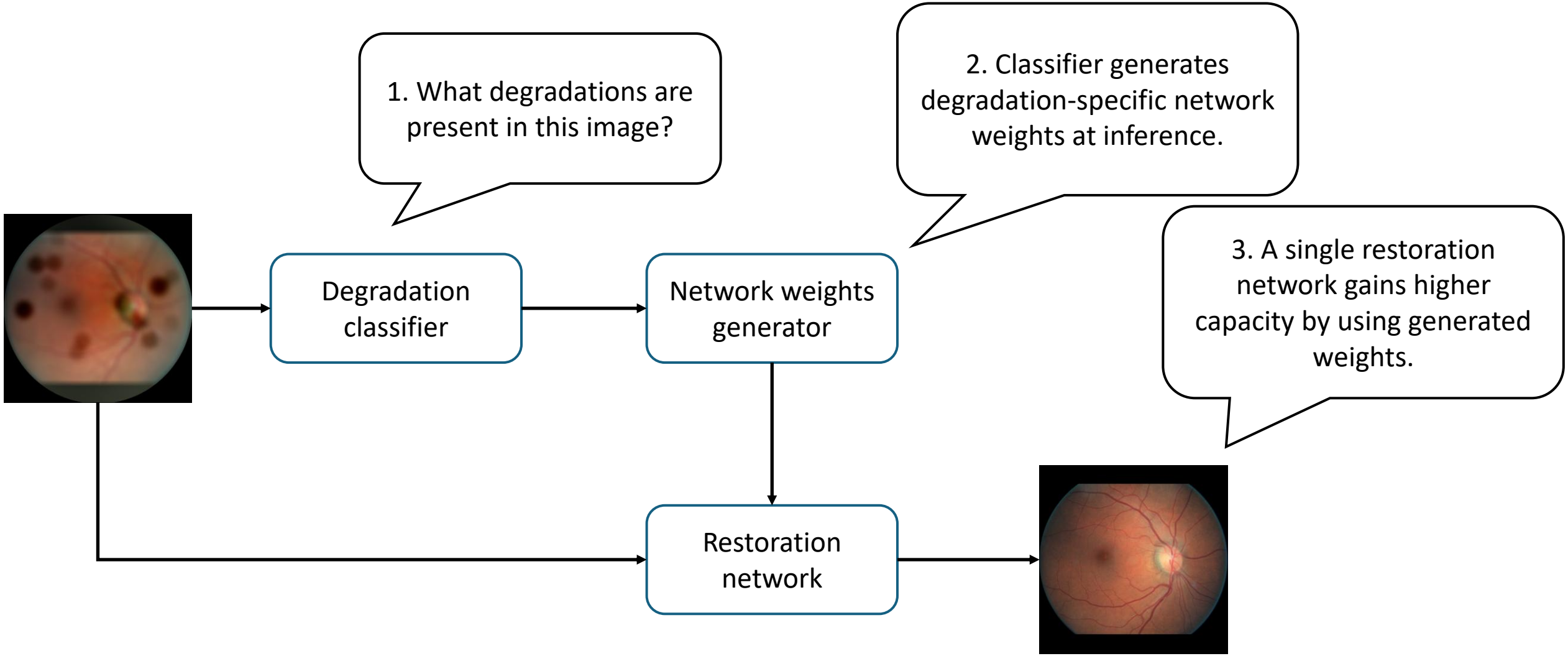
Collecting paired training data (low-light and normal-light images) can be time-consuming!

Our solution removes the need for ground-truth data when training a low-light enhancement network!



- Challenge: Networks must handle diverse and even coexisting degradations!
- A single model struggles because:
  - Artefacts affect local areas (additive)
  - Illumination affects global channels (multiplicative)
- Using multiple models is often too heavy

Top: degraded images Bottom: high-quality images



Our solution to train only one network but can generate weights for different input degradations!